

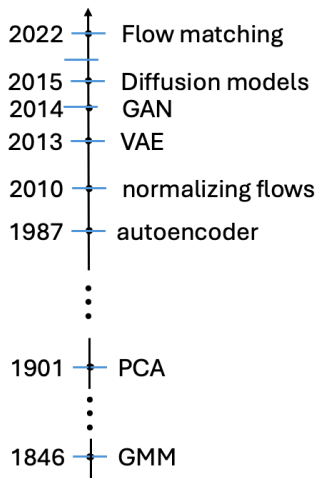
COMM114: Generative AI - Normalizing Flows

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September 2025

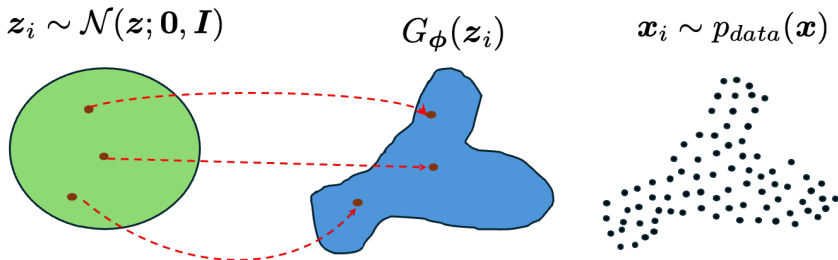
Normalizing Flows

Normalizing flow (NF) was invented in 2010.



Recap

- GAN learns a generator G_ϕ that maps a Gaussian noise $z \sim \mathcal{N}(z; \mathbf{0}, \mathbf{I})$ to a sample $G_\phi(z)$ that looks as if it came from the data distribution $p_{data}(\mathbf{x})$.
- A discriminator D_θ is introduced to help learning the generator G_ϕ .
- In the ideal case, the model distribution $p_\phi(\mathbf{x})$ coincides with the data distribution $p_{data}(\mathbf{x})$.



Demonstration of NF



Picture from "Glow: Generative Flow with Invertible 1×1 Convolutions" by Diederik P. Kingma and Prafulla Dhariwal.

Objective of NF

- **Input:** data samples $\{x_i\}_{i=1}^n$.



- **Assumption:** Assume that the data samples are drawn from a certain data distribution $p_{data}(\cdot)$.
- **Objective of NF:** To learn an **invertible and differentiable** function $f_{\theta}(\cdot)$ (e.g., a deep neural network) that maps a Gaussian noise $z \sim \mathcal{N}(z; \mathbf{0}, \mathbf{I})$ to a sample $f_{\theta}(z)$ that looks as if it came from the data distribution $p_{data}(x)$.
- Unlike GAN, no discriminator is introduced in NF.
- **The invertibility and differentiability of $f_{\theta}(\cdot)$ makes the learning process much easier.**
- To facilitate invertibility of f_{θ} , the input and output of $f_{\theta}(\cdot)$ are set to the same dimension.

Explicit PDF with Functional Invertibility

- **Definition:** a function $f_{\theta}(\cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is invertible if it is a one-to-one mapping.
- Denote its inverse function as $h_{\theta}(\cdot)$. We have

$$\mathbf{z} = h_{\theta}(\mathbf{x}) \text{ for every } \mathbf{x} = f_{\theta}(\mathbf{z}) \text{ where } \mathbf{z} \sim \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I}).$$

- Furthermore, if the function $f_{\theta}(\cdot)$ is differentiable. Then $h_{\theta}(\cdot)$ is also differentiable. The Jacobian matrix of $h_{\theta}(\cdot)$

$$Dh_{\theta}(\mathbf{x}) = \frac{dh_{\theta}(\mathbf{x})}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial h_{\theta}(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial h_{\theta}(\mathbf{x})}{\partial x_d} \\ \vdots & & \vdots \\ \frac{\partial h_{\theta,d}(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial h_{\theta,d}(\mathbf{x})}{\partial x_d} \end{bmatrix} = \begin{bmatrix} \frac{\partial h_{\theta,1}(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial h_{\theta,1}(\mathbf{x})}{\partial x_d} \\ \vdots & & \vdots \\ \frac{\partial h_{\theta,d}(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial h_{\theta,d}(\mathbf{x})}{\partial x_d} \end{bmatrix} \quad (1)$$

is invertible. Its determinant is nonzero, i.e., $\det(Dh_{\theta}(\mathbf{x})) \neq 0$.

Explicit PDF with Functional Invertibility (cont.)

Explicit PDF for $x = f_{\theta}(z)$

Suppose $f_{\theta}(\cdot)$ is differentiable and invertible. Its inversion function $h_{\theta}(\cdot)$ is also differentiable and invertible. The probability density function (PDF) of $x = f_{\theta}(z)$ takes an explicit form:

$$p_X(x; \theta) = p_Z(h_{\theta}(x)) |\det Dh_{\theta}(x)|, \quad (2)$$

where $\det(Dh_{\theta}(x))$ is the determinant of the Jacobian matrix $Dh_{\theta}(x)$ given by (1). The operation $|\cdot|$ denotes the absolute value.

Remark: The conclusion above is in fact obtained by applying "change variables formula".

Exercise 1

- What is the determinant of the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$?
- What is the determinant of the matrix $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$?

Exercise 2

- What is the Jacobian matrix of

$$\mathbf{y} = A\mathbf{x}$$
$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}?$$

Determinant of an upper (or lower) triangle matrix

- Suppose $A = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ a_{21} & a_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{(n-1)1} & a_{(n-2)2} & \dots & 0 \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$.
- Its determinant is

$$\det(A) = \prod_{i=1}^n a_{ii}$$

Training via Maximum log-likelihood

Explicit PDF for $\mathbf{x} = f_{\boldsymbol{\theta}}(\mathbf{z})$ makes the training process much easier

- Input: data samples $\{\mathbf{x}_i\}_{i=1}^n$
- The parameter $\boldsymbol{\theta}$ can be learned by maximizing the log-likelihood via

$$\begin{aligned} & \arg \max_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^n \log(p_{\mathbf{X}}(\mathbf{x}_i; \boldsymbol{\theta})) \\ &= \arg \max_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^n \left(\log p_{\mathbf{Z}}(h_{\boldsymbol{\theta}}(\mathbf{x}_i)) + \log |\det D h_{\boldsymbol{\theta}}(\mathbf{x}_i)| \right), \end{aligned}$$

where the expression for $p_{\mathbf{X}}^{\boldsymbol{\theta}}(\mathbf{x})$ follows directly from (2).

Background on "change of variables formula"

- Let X be a continuous scalar random variable.
- The cumulative density function (CDF) of X is $F_X(a) = P(X \leq a)$
- The probability density function (pdf) of X is $p_X(a) = F'_X(a) = \frac{dF_X(a)}{da}$.
- Typical scalar random variables:
 - Gaussian: $X \sim \mathcal{N}(\mu, \sigma)$ where $p_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$
 - Uniform: $X \sim \mathcal{U}(a, b)$, where $p_X(x) = \frac{1}{b-a} 1_{a \leq x \leq b}$

Background on "change of variables formula" (cont.)

Exercise:

- Let Z be a uniform random variable with the probability density function (PDF) $p_Z = 1_{0 \leq z \leq 1}$. What is $p_Z(0.5)$?
 - Answer: $p_Z(0.5) = 1$.
- Let $X = 5Z$, and let p_X be its PDF. What is $p_X(2.5)$?
 - Answer: X has a uniform distribution over $[0, 5]$. So $p_X(2.5) = \frac{1}{5}$.

Question

What is the general rule for computing PDF for the continuous scalar variable $X = f(Z)$?

Answer: To apply "change of variables formula".

Background on "change of variables formula" (cont.)

Change of variables for 1D case

- If $X = f(Z)$ and $f(\cdot)$ is monotone with inverse $Z = f^{-1}(X) = h(X)$, then we have:

$$p_X(x) = p_Z(h(x))|h'(x)|.$$

- Proof: Assume $f(\cdot)$ is monotonically increasing

$$F_X(x) = p(X \leq x) = p[f(Z) \leq x] = p[Z \leq h(x)] = F_Z(h(x))$$

Taking derivatives on both sides, we have:

$$p_X(x) = \frac{dF_X(x)}{dx} = \frac{dF_Z(h(X))}{dx} = P_Z(h(x))h'(x)$$

Alternative Expression

$p_X(x)$ can be alternatively expressed as

$$p_X(x) = p_Z(h(x)) \frac{1}{|f'(z)|}$$

by using the fact that $h'(x) = \frac{1}{f'(z)}$.

Background on "change of variables formula" (cont.)

Change of variables for high-dimensional case

- Let $\mathbf{Z}$ be a uniform random vector in the d -dimensional cube $[0, 1]^d$.
- Let $\mathbf{X} = \mathbf{AZ}$, where A is a square invertible matrix, with inverse $B = A^{-1}$. How is $\mathbf{X}$ distributed?
 - $\mathbf{X}$ is uniformly distributed over a parallelotope of volume $|\det(A)|$.

$$\begin{aligned} p_{\mathbf{X}}(\mathbf{x}) &= p_{\mathbf{Z}}(B\mathbf{x})|\det(B)| \\ &= p_{\mathbf{Z}}(B\mathbf{x})\frac{1}{|\det(A)|}. \end{aligned}$$

Remark: It is known from algebra that if $B = A^{-1}$, then $\det(B) = \frac{1}{\det(A)}$.

Background on "change of variables formula" (cont.)

- For a linear mapping specified via $\mathbf{X} = \mathbf{A}\mathbf{Z}$, change in volume is given by the determinant of $\mathbf{A}$.
- For a non-linear mapping $\mathbf{f}(\cdot)$, the linearized change in volume is given by the determinant of the Jacobian of $\mathbf{f}(\cdot)$.

Explicit PDF for $\mathbf{x} = \mathbf{f}_\theta(\mathbf{z})$

Suppose $\mathbf{f}_\theta(\cdot)$ is differentiable and invertible. Its inversion function $\mathbf{h}_\theta(\cdot)$ is also differentiable and invertible. The probability density function (PDF) of $\mathbf{x} = \mathbf{f}_\theta(\mathbf{z})$ takes an explicit form:

$$\begin{aligned} p_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) &= p_{\mathbf{Z}}(\mathbf{h}_\theta(\mathbf{x})) |\det D\mathbf{h}_\theta(\mathbf{x})| \\ &= p_{\mathbf{Z}}(\mathbf{h}_\theta(\mathbf{x})) \frac{1}{|\det D\mathbf{f}_\theta(\mathbf{x})|} \end{aligned}$$

where $\det D(\mathbf{h}_\theta(\mathbf{x}))$ is the determinant of the Jacobian matrix $D\mathbf{h}_\theta(\mathbf{x})$ given by (1). The operation $|\cdot|$ denotes the absolute value.

NF

In normalizing flow (NF), an explicit probability density function (PDF) can be obtained after applying an invertible mapping.

- Input: data samples $\{\mathbf{x}_i\}_{i=1}^n$
- Let $\mathbf{z} \sim \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$.
- Let $f_{\boldsymbol{\theta}} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be differentiable and invertible. Its inverse function is $h_{\boldsymbol{\theta}}(\cdot) = f_{\boldsymbol{\theta}}^{-1}(\cdot)$.
- The parameter $\boldsymbol{\theta}$ in $f_{\boldsymbol{\theta}}$ can be learned by maximizing the log-likelihood via

$$\begin{aligned} & \arg \max_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^n \log(p_{\mathbf{X}}^{\boldsymbol{\theta}}(\mathbf{x}_i)) \\ &= \arg \max_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^n \left(\log p_{\mathbf{Z}}(h_{\boldsymbol{\theta}}(\mathbf{x}_i)) + \log |\det D h_{\boldsymbol{\theta}}(\mathbf{x}_i)| \right) \\ &= \arg \max_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^n \left(\log p_{\mathbf{Z}}(h_{\boldsymbol{\theta}}(\mathbf{x}_i)) - \log |\det D f_{\boldsymbol{\theta}}(\mathbf{x}_i)| \right) \end{aligned}$$

Composition of Flows

- To facilitate the design of f_{θ} in NF, x is computed as

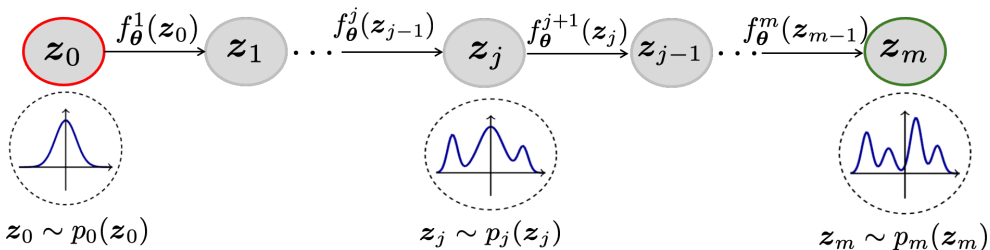
$$\begin{aligned}x &= f_{\theta}^m \circ \cdots \circ f_{\theta}^1(z) \\&= f_{\theta}^m(f_{\theta}^{m-1}(\cdots(f_{\theta}^1(z)))) \\&= f_{\theta}(z).\end{aligned}$$

- The overall function f_{θ} consists of a sequence of m invertible mappings to obtain x .
- Each mapping $f_{\theta}^j(\cdot)$ can be a relatively simple function.

Composition of Flows (cont.)

Probability path over mapping index j :

- As the mapping index j increases from $j = 0$ until $j = m$, the probability density function $p_j(z_j)$ gradually transforms from a Gaussian distribution $p_0(z_0)$ to an approximation $p_m(x)$ (we let $x = z_m$) of the data distribution $p_{data}(x)$.



- As a result, we obtain a probability path $p_j(z_j)$ over the mapping index $0 \leq j \leq m$.

Composition of Flows (cont.)

- The Jacobian matrix of a composite function is a product of those individual Jacobian matrices:

$$\frac{\partial(f_{\boldsymbol{\theta}})^{-1}(\boldsymbol{x})}{\partial\boldsymbol{x}} = \prod_{j=1}^m \left(\frac{\partial(f_{\boldsymbol{\theta}}^j)^{-1}(\boldsymbol{z}_j)}{\partial\boldsymbol{z}_j} \right) \quad [\boldsymbol{z}_m = \boldsymbol{x}]$$

- The PDF $p_{\boldsymbol{X}}(\boldsymbol{x}; \boldsymbol{\theta})$ for a composition of flows is given by

$$p_{\boldsymbol{X}}(\boldsymbol{x}; \boldsymbol{\theta}) = p_{\boldsymbol{Z}}(\boldsymbol{f}_{\boldsymbol{\theta}}^{-1}(\boldsymbol{x})) \left| \det \prod_{j=1}^m \left(\frac{\partial(f_{\boldsymbol{\theta}}^j)^{-1}(\boldsymbol{z}_j)}{\partial\boldsymbol{z}_j} \right) \right|$$

Composition of Flows (cont.)

- Property: Determinant of product equals product of determinant.

$$\begin{aligned} p_X(\mathbf{x}; \boldsymbol{\theta}) &= p_Z(\mathbf{f}_{\boldsymbol{\theta}}^{-1}(\mathbf{x})) \left| \det \prod_{j=1}^m \left(\frac{\partial (f_{\boldsymbol{\theta}}^j)^{-1}(\mathbf{z}_j)}{\partial \mathbf{z}_j} \right) \right| \\ &= p_Z(\mathbf{f}_{\boldsymbol{\theta}}^{-1}(\mathbf{x})) \prod_{j=1}^m \left| \det \underbrace{\left(\frac{\partial (f_{\boldsymbol{\theta}}^j)^{-1}(\mathbf{z}_j)}{\partial \mathbf{z}_j} \right)}_{Dh_{\boldsymbol{\theta}}^j(\mathbf{z}_j)} \right| \end{aligned}$$

Training of Compositions of Flows

- Input: data samples $\{\mathbf{x}_i\}_{i=1}^n$
- Let $\mathbf{z} \sim \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$.
- Let $f_{\boldsymbol{\theta}}(\mathbf{z}) = f_{\boldsymbol{\theta}}^m \circ \dots \circ f_{\boldsymbol{\theta}}^1(\mathbf{z})$
 - Denote the inverse function of each individual function $f_{\boldsymbol{\theta}}^j(\mathbf{z})$ as $h_{\boldsymbol{\theta}}^j(\mathbf{z})$
- The parameter $\boldsymbol{\theta}$ in $f_{\boldsymbol{\theta}}$ can be learned by maximizing the log-likelihood via

$$\begin{aligned} & \arg \max_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^n \log(p_{\mathbf{X}}^{\boldsymbol{\theta}}(\mathbf{x}_i)) \\ &= \arg \max_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^n \left(\log p_{\mathbf{Z}}(h_{\boldsymbol{\theta}}(\mathbf{x}_i)) + \sum_{j=1}^m \log |\det D h_{\boldsymbol{\theta}}^j(\mathbf{z}_{ij})| \right) \quad [\mathbf{z}_{im} = \mathbf{x}_i] \\ &= \arg \max_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^n \left(\log p_{\mathbf{Z}}(h_{\boldsymbol{\theta}}(\mathbf{x}_i)) - \sum_{j=1}^m \log |\det D f_{\boldsymbol{\theta}}^j(\mathbf{z}_{ij})| \right) \end{aligned}$$

Suppose the NF model is well trained. We can perform the following operations exactly.

- Exact likelihood evaluation for computing $\log p_{\mathbf{X}}(\mathbf{x}_i; \boldsymbol{\theta})$
- Sampling for generating new data

$$\begin{aligned} \mathbf{x} &= \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{z}) \text{ where } \mathbf{z} \sim p_0(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I}) \\ &= \mathbf{f}_{\boldsymbol{\theta}}^m(\mathbf{f}_{\boldsymbol{\theta}}^{m-1}(\cdots(\mathbf{f}_{\boldsymbol{\theta}}^1(\mathbf{z})))) \end{aligned}$$

Summary

PGMs	GMMs	VAEs	GANs	NFs	Diffusion Models
Quality			High		
Sampling	Easy	Easy	Easy		
Evaluating $p_{\theta}(x)$	Easy	approximate evaluation	Not possible		

References

- Lecture notes by Stefano Ermon
- Youtube talk by Marcus Brubaker

Questions?